

CHAPTER 3: PLANT KINGDOM

Complete, Exam-Oriented, High-Yield Educational Master Document

Core Taxonomic Categories Covered:

Algae • Bryophytes • Pteridophytes • Gymnosperms • Angiosperms

CHAPTER MIND MAP — TAXONOMIC HIERARCHY

1. ALGAE

Simple, thalloid, autotrophic, aquatic.

● Chlorophyceae (Green)

♥ Phaeophyceae (Brown)

● Rhodophyceae (Red)

2. BRYOPHYTES

Amphibians of Plant Kingdom; non-vascular.

🍃 Liverworts (Marchantia)

🌱 Mosses (Funaria, Sphagnum)

3. PTERIDOPHYTES

First vascular plants; seedless; homosporous/heterosporous.

♦ Psilopsida • Lycopsida

4. GYMNOSPERMS

Naked seeds; ovules exposed without ovary wall.

🌲 **Conifers & Cycads**

📖 **Examples:** Cycas, Pinus, Ginkgo

5. ANGIOSPERMS

Flowering plants; enclosed seeds in fruits.

🌱 **Dicotyledons** (2 cotyledons)

🌾 **Monocotyledons** (1 cotyledon)

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3.0 Historical & Modern Systems of Plant Classification

● 1. Artificial Classification Systems

- **Basis:** Based on gross superficial morphological characters such as habit, colour, number, and shape of leaves.
- **Linnaeus System:** Based mainly on vegetative characters or on the **androecium structure**.
- **Major Drawback:** Separated closely related species because it relied on only a few characteristics.
- **Critical Defect:** Gave equal weightage to vegetative and sexual characteristics. This is unacceptable because vegetative characters are easily affected by environment.

● 2. Natural Classification Systems

- **Basis:** Based on natural affinities among organisms. Considers both external and internal features.
- **Internal Features Examined:** Ultrastructure, anatomy, embryology, and phytochemistry.
- **Proposed By:** **George Bentham** and **Joseph Dalton Hooker** for flowering plants.

● 3. Phylogenetic Classification Systems

- **Basis:** Based on **evolutionary relationships** between organisms.
- **Core Premise:** Organisms belonging to the same taxa have a **common ancestor**. Currently widely accepted by modern taxonomists.

● 4. Modern Taxonomic Methods (Fossil-Free)

- **Numerical Taxonomy:** Carried out using computers based on all observable characters. Numbers and codes are assigned to characters. Gives equal importance to hundreds of characters simultaneously.
- **Cytotaxonomy:** Based on cytological information such as **chromosome number, structure, and behaviour**.
- **Chemotaxonomy:** Uses the chemical constituents of plants to resolve taxonomic confusions.

General Characteristics of Algae

Algae are **chlorophyll-bearing, simple, thalloid, autotrophic** and largely **aquatic** (both fresh water and marine) organisms. They occur in various habitats: moist stones, soils, and wood. Some occur in association with fungi (**lichens**) and animals (e.g., on **sloth bear**).

Form & Size Variation:

- Colonial: *Volvox*
- Filamentous: *Ulothrix*, *Spirogyra*
- Massive plant bodies: Marine Kelps (up to 100 meters)

Reproduction Modes:

- **Vegetative:** By fragmentation.
- **Asexual:** Flagellated zoospores (motile).
- **Sexual:** Fusion of two gametes.

Sexual Gamete Types:

- **Isogamous:** Flagellated (*Ulothrix*) or Non-flagellated (*Spirogyra*).
- **Anisogamous:** Dissimilar in size (*Eudorina*).
- **Oogamous:** Large static female + small motile male (*Volvox*, *Fucus*).

Economic & Ecological Importance of Algae

- **Carbon Fixation:** Perform at least **half (50%) of the total CO₂ fixation** on Earth through photosynthesis.
- **Oxygen Producers:** Increase the level of dissolved oxygen in aquatic ecosystems.
- **Primary Producers:** Form the foundation of aquatic food cycles.
- **Food Source:** ~70 species of marine algae used as food, e.g., *Porphyra*, *Laminaria*, *Sargassum*.

- **Hydrocolloids (Water-holding substances):**

- **Algin:** Produced by brown algae.

- **Carrageen:** Produced by red algae.

- **Agar:** Commercial product obtained from *Gelidium* and *Gracilaria*; used to grow microbes and in ice-creams/jellies.

- **Space Food:** *Chlorella* (unicellular, rich in proteins) used as food supplement by space travellers.

3.1.1 Chlorophyceae (Green Algae)

Habit/Form: Unicellular, colonial, or filamentous. Grass green due to **Chlorophyll a & b**.

Chloroplast Shapes: Discoid, plate-like, reticulate, cup-shaped, spiral, or ribbon-shaped.

Storage Bodies: Pyrenoids located in chloroplasts (contain protein besides starch). May also store oil droplets.

Cell Wall: Rigid wall with inner layer of **cellulose** and outer layer of **pectose**.

Reproduction: Vegetative (fragmentation), Asexual (flagellated zoospores in zoosporangia), Sexual (isogamous, anisogamous, oogamous).

Examples: Chlamydomonas, Volvox, Ulothrix, Spirogyra, Chara.

3.1.2 Phaeophyceae (Brown Algae)

Habitat: Primarily marine. Range from simple branched (*Ectocarpus*) to massive kelps (100 m).

Pigments: Chlorophyll a, c, carotenoids, xanthophylls (**fucoxanthin** gives olive-green to brown colour).

Stored Food: Complex carbohydrates as **laminarin or mannitol**.

Structure: Cellulosic wall with outer gelatinous coating of **algin**. Plant body attached by **holdfast**, stalk (**stipe**), and leaf-like photosynthetic organ (**frond**).

Gametes/Spores: Biflagellate zoospores/gametes are **pyriform (pear-shaped)** with 2 unequal lateral flagella.

Examples: Ectocarpus, Dictyota, Laminaria, Sargassum, Fucus.

3.1.3 Rhodophyceae (Red Algae)

Habitat: Mostly marine (warmer areas). Occur near surface and at great ocean depths.

Pigment: Red colour due to predominance of **r-phycoerythrin** (also Chlorophyll a & d).

Stored Food: **Floridean starch** (structural similarity to **amylopectin** and **glycogen**).

Cell Wall: Cellulose, pectin, and polysulphate esters.

Reproduction: Asexual by **non-motile spores**; Sexual by **non-motile gametes** (Oogamous with complex post-fertilisation developments).

Examples: Polysiphonia, Porphyra, Gracilaria, Gelidium.

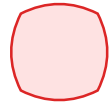
TABLE 3.1 — DIVISIONS OF ALGAE AND THEIR MAIN CHARACTERISTICS (NCERT PAGE 27)

Classes	Common Name	Major Pigments	Stored Food	Cell Wall	Flagellar Number & Position	Habitat
Chlorophyceae	Green algae	Chlorophyll a, b	Starch	Cellulose	2–8, equal, apical	Fresh water, brackish water, salt water
Phaeophyceae	Brown algae	Chlorophyll a, c, fucoxanthin	Mannitol, laminarin	Cellulose and algin	2, unequal, lateral	Fresh water (rare), brackish water, salt water
Rhodophyceae	Red algae	Chlorophyll a, d, phycoerythrin	Floridean starch	Cellulose, pectin & polysulphate esters	Absent (Non-motile)	Fresh water (some), brackish water, salt water (most)

FIGURE 3.1 — REPRESENTATIVE ALGAE STRUCTURAL SCHEMATICS (NCERT FIGURE 3.1)



(Brown: Frond, Stipe, Holdfast)



Porphyra

(Red: Flat Thallus)

(Brown: Air Bladders)



Polysiphonia

(Red: Branched Tree)

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3.2 Division Bryophytes (Amphibians of the Plant Kingdom)

Essential Characteristics

- **Amphibians of Plant Kingdom:** Can live in soil but dependent on **water for sexual reproduction** (transfer of antherozoids).
- **Habitat:** Damp, humid, and shaded localities in hills.
- **Ecological Role:** Major role in plant succession on bare rocks/soil.
- **Body Differentiation:** Thallus-like (prostrate or erect), attached to substrate by unicellular or multicellular **rhizoids**. Lack true roots, stems, or leaves (possess root-like, leaf-like, or stem-like structures).

Life Cycle & Reproduction

- **Dominant Phase:** Haploid **gametophyte** (produces gametes).
- **Male Sex Organ: Antheridium** → produces biflagellate *antherozoids*.
- **Female Sex Organ: Archegonium** (flask-shaped) → produces a single *egg*.
- **Sporophyte:** Multicellular, **not free-living**; attached to photosynthetic gametophyte for nourishment. Undergoes meiosis in capsule to produce haploid spores.

★ Economic & Ecological Importance of Bryophytes

- **Peat Production:** Species of *Sphagnum* (a moss) provide **peat** used as fuel and as packing material for trans-shipment of living material due to high water-holding capacity.
- **Rock Colonization & Succession:** Mosses + lichens decompose rocks, creating fertile substrate for higher plants.
- **Soil Conservation:** Mosses form dense mats on soil, reducing rain impact and preventing soil erosion.

3.2.1 Liverworts (e.g., *Marchantia*)

- **Plant Body:** Thalloid, **dorsiventral**, closely appressed to substrate. Leafy members have tiny leaf-like appendages in 2 rows on stem-like structures.
- **Asexual Reproduction:**
 1. Fragmentation of thalli.
 2. **Gemmae:** Green, multicellular, asexual buds developed in **gemma cups** on thalli. Detach and germinate into new individuals.
- **Sexual Reproduction:** Sex organs produced on same or different thalli (Antheridiophore / Archegoniophore).
- **Sporophyte Structure:** Differentiated into **Foot, Seta, and Capsule**. Spores formed after meiosis inside capsule germinate into free-living gametophytes.

3.2.2 Mosses (e.g., *Funaria*, *Polytrichum*, *Sphagnum*)

- **Dominant Gametophyte Stages:**
 1. **Protonema Stage:** Creeping, green, branched, filamentous stage developing directly from spore.
 2. **Leafy Stage:** Develops from secondary protonema as lateral bud; upright axis with spirally arranged leaves & multicellular branched rhizoids. Bears sex organs.
- **Reproduction:** Vegetative by fragmentation & budding in secondary protonema. Sex organs (antheridia & archegonia) at apex of leafy shoots.
- **Sporophyte:** Consists of foot, seta, capsule. More elaborate than liverworts; features an **elaborate spore dispersal mechanism**.

- Includes **horsetails and ferns**.
- First terrestrial plants to possess **vascular tissues (Xylem & Phloem)**.
- Used for medicinal purposes, soil-binders, and ornamentals.
- Inhabit cool, damp, shady places.

Dominant Sporophyte Body

- Main plant body is a **Sporophyte (2n)** differentiated into true root, stem, leaves.
- **Microphylls**: Small leaves (e.g., *Selaginella*).
- **Macrophylls**: Large leaves (e.g., Ferns).
- Sporangia subtended by **sporophylls**; may form compact **strobili or cones** (*Selaginella*, *Equisetum*).

Gametophyte & Fertilisation

- Spores germinate to form **Prothallus** (small, inconspicuous, free-living, photosynthetic thalloid gametophyte).
- Requires water for transfer of male antherozoids to archegonium.
- Restricted geographical distribution due to water requirement.

EVOLUTIONARY SIGNIFICANCE: Heterospory & Precursor to Seed Habit

Homosporous vs. Heterosporous:

- **Homosporous**: Majority of pteridophytes produce spores of similar kinds.
- **Heterosporous**: Genera like *Selaginella* and *Salvinia* produce two kinds of spores — **Megaspores** (large) and **Microspores** (small).

Precursor to Seed Habit (Crucial Exam Concept):

- Megaspores & microspores germinate into female & male gametophytes.
- **Female gametophyte is retained on parent sporophyte** for variable periods.
- Development of zygote into embryo takes place within female gametophyte. This event is a **precursor to the seed habit**.

TAXONOMIC CLASSIFICATION OF PTERIDOPHYTES (4 CLASSES)

1. Psilopsida

Example: *Psilotum*

2. Lycopsidea

Examples: *Selaginella*, *Lycopodium*

3. Sphenopsida

4. Pteropsida

Example: *Equisetum*

Examples: *Dryopteris*, *Pteris*, *Adiantum*

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3.4 Division Gymnosperms (Naked-Seeded Plants)

Definition & Habit

- **Gymnosperms (gymnos: naked, sperma: seeds):** Ovules are **not enclosed by any ovary wall** and remain exposed before & after fertilisation.
- Seeds post-fertilisation are naked (not covered).
- Includes tall trees, medium-sized trees, and shrubs.
- **Sequoia (Giant Redwood):** One of the tallest tree species.

Morphological Adaptations

- **Roots:** Tap roots.
 - Fungal association = **Mycorrhiza** (*Pinus*).
 - Specialized roots = **Coralloid roots** with N₂-fixing cyanobacteria (*Cycas*).
- **Stems:** Unbranched (*Cycas*) or Branched (*Pinus*, *Cedrus*).
- **Leaves:** Pinnate leaves persist in *Cycas*. Conifers have needle-like leaves, thick cuticle, and **sunken stomata** to reduce water loss.

Reproduction & Cones

- **Heterosporous:** Produce haploid microspores & megaspores.
- **Male Strobili (Microsporangiate):** Microsporophylls + Microsporangia → **Pollen Grains** (reduced male gametophyte).
- **Female Strobili (Macrosporangiate):** Megasporophylls + Ovules (Megasporangia).
- Monoecious (*Pinus*: both cones on same tree); Dioecious (*Cycas*: male cone & megasporophylls on different trees).

STEP-BY-STEP FERTILISATION & EMBRYO DEVELOPMENT IN GYMNOSPERMS

Megaspore Mother Cell undergoes meiosis → 4 Megaspores (1 survives in Nucellus)



Megaspore forms Multicellular Female Gametophyte with 2+ Archegonia



Pollen grain carried by air currents to ovule opening



Pollen tube grows towards archegonia → fertilisation → Zygote → Embryo & Naked Seed

3.5 Division Angiosperms (Flowering Plants)

General Characteristics

- **Enclosed Seeds:** Pollen grains and ovules developed in specialized structures called **flowers**. Seeds are enclosed inside **fruits**.
- **Size Range:** Extremely wide — from microscopic *Wolffia* (smallest) to tall *Eucalyptus* trees (>100 meters).
- **Economic Value:** Provide food, fodder, fuel, medicines, and commercial products.

Two Main Classes

Dicotyledons:

- 2 Cotyledons in seeds
- Reticulate venation
- Tetramerous/Pentamerous flowers

Monocotyledons:

- 1 Cotyledon in seed
- Parallel venation
- Trimerous flowers

Feature	Algae	Bryophytes	Pteridophytes	Gymnosperms	Angiosperms
Dominant Phase	Gametophyte (n)	Gametophyte (n)	Sporophyte (2n)	Sporophyte (2n)	Sporophyte (2n)
Vascular Tissue	Absent	Absent	Present (Xylem/Phloem)	Present	Present
Plant Body Differentiation	Thalloid	Thallus / Stem & Leaf-like	True Root, Stem, Leaves	True Root, Stem, Leaves	True Root, Stem, Leaves
Water Needed for Fertilisation	Yes	Yes ("Amphibians")	Yes	No (Pollen tube / Air)	No (Pollen tube / Vectors)
Seed Formation	Absent	Absent	Absent (Precursor habit)	Naked Seeds	Enclosed in Fruit



Memory Tricks & High-Yield Exam Points

💡 Brown Algae Examples Mnemonic:

"Self-Help Group: S-E-L-F-D"

- S: *Sargassum*
- E: *Ectocarpus*
- L: *Laminaria*
- F: *Fucus*
- D: *Dictyota*

💡 Red Algae Examples Mnemonic:

"P-G-P-G Square"

P: *Polysiphonia*

G: *Gracilaria*

P: *Porphyra*

G: *Gelidium*

★ High-Yield Exam Reminders:

- **Floridean Starch** is structurally similar to **amylopectin and glycogen**.
- **Heterosporous Pteridophytes:** *Selaginella* & *Salvinia*.
- **Mycorrhiza** in *Pinus*; **Coralloid roots** in *Cycas*.
- **Smallest Angiosperm:** *Wolffia*.

FINAL REVISION SUMMARY (Quick Exam Recap)

- ✓ **Whittaker 5-Kingdom:** Monera, Protista, Fungi, Animalia, Plantae. Cyanobacteria are BGA (not true algae).
- ✓ **Algae Classes:** Chlorophyceae (green, starch), Phaeophyceae (brown, mannitol/laminarin), Rhodophyceae (red, floridean starch).
- ✓ **Bryophytes:** Non-vascular land plants; liverworts (gemma cups) & mosses (protonema stage).
- ✓ **Pteridophytes:** First vascular plants (Xylem/Phloem); sporophyte dominant; seed habit precursor in heterosporous species.
- ✓ **Gymnosperms:** Naked ovules/seeds; needle leaves with sunken stomata; reduced pollen grain gametophyte.
- ✓ **Angiosperms:** Seeds enclosed in fruits; double fertilisation; divided into Dicots & Monocots.

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Complete NCERT Exercise Solutions

Q1: What is the basis of classification of algae?

Answer: Algae are classified into three main classes (Chlorophyceae, Phaeophyceae, Rhodophyceae) based on:

1. **Nature of Photosynthetic Pigments:** Chlorophyll a, b in Chlorophyceae; Chlorophyll a, c & fucoxanthin in Phaeophyceae; Chlorophyll a, d & r-phycoerythrin in Rhodophyceae.
2. **Type of Stored Food:** Starch in Chlorophyceae; Laminarin/Mannitol in Phaeophyceae; Floridean starch in Rhodophyceae.
3. **Cell Wall Composition:** Inner cellulose + outer pectose in green algae; Cellulose + algin in brown algae; Cellulose + pectin + polysulphate esters in red algae.
4. **Flagellar Number and Position:** 2–8 equal apical flagella in green algae; 2 unequal lateral flagella in brown algae; Flagella absent in red algae.

Q2: When and where does reduction division take place in the life cycle of a liverwort, a moss, a fern, a gymnosperm and an angiosperm?

Answer:

- **Liverwort:** Takes place in the *capsule of the sporophyte* in spore mother cells to form haploid spores.
- **Moss:** Takes place in the *capsule of the sporophyte* in spore mother cells prior to spore dispersal.
- **Fern:** Takes place inside *sporangia (borne on sporophylls)* in spore mother cells.
- **Gymnosperm:** Takes place in *microsporangia* (for microspores) and in the *nucellus/megasporangium* (in megaspore mother cell).
- **Angiosperm:** Takes place in *anthers* (microspore mother cells) and *ovules* (megaspore mother cells).

Q3: Name three groups of plants that bear archegonia. Briefly describe the life cycle of any one of them.

Answer: The three groups that bear archegonia are **Bryophytes, Pteridophytes, and Gymnosperms.**

Life Cycle of Bryophytes (Funaria/Moss): Alternation of generations occurs between a dominant haploid gametophyte (n) and a dependent diploid sporophyte (2n). Spores germinate to form a creeping filamentous *protonema*, which gives rise to an erect leafy stage bearing antheridia & archegonia. Water carries biflagellate antherozoids to archegonia to form a diploid zygote. The zygote develops into a sporophyte (foot, seta, capsule), where meiosis produces haploid spores.

Q4: Mention the ploidy of the following:

Protonemal cell of a moss: **Haploid (n)**

Primary endosperm nucleus in dicot: **Triploid (3n)**

Leaf cell of a moss: **Haploid (n)**

Prothallus cell of a fern: **Haploid (n)**

Gemma cell in Marchantia: **Haploid (n)**

Meristem cell of monocot: **Diploid (2n)**

Ovum of a liverwort: **Haploid (n)**

Zygote of a fern: **Diploid (2n)**

Q5: Write a note on economic importance of algae and gymnosperms.

Economic Importance of Algae: Carbon dioxide fixation (50% on Earth), primary producers in aquatic food webs, hydrocolloids (algin, carrageen), commercial agar from *Gelidium/Gracilaria*, food sources (*Porphyra, Laminaria*), protein supplement (*Chlorella*).

Economic Importance of Gymnosperms: Timber and wood products (*Pinus, Cedrus*), resins and turpentine, ornamental trees (*Cycas, Ginkgo*), edible seeds (*Pinus gerardiana* - chilgoza).

Q6: Both gymnosperms and angiosperms bear seeds, then why are they classified separately?

Answer: Gymnosperms have **naked seeds** because their ovules are not enclosed by an ovary wall before or after fertilisation. Angiosperms have **enclosed seeds** contained inside fruits formed from ovaries. Furthermore, angiosperms produce true flowers and undergo **double fertilisation**, whereas gymnosperms lack true flowers and fruit formation.

Q7: What is heterospory? Briefly comment on its significance. Give two examples.

Answer: Heterospory is the production of two morphologically distinct types of spores in the same plant: smaller **microspores** (male) and larger **megaspores** (female).

Significance: The megaspore develops into a female gametophyte that is retained inside the parent sporophyte. This provides nourishment and protection during embryo development, serving as an essential **precursor to the seed habit** in plant evolution.

Examples: *Selaginella* and *Salvinia*.

Q8: Explain briefly the following terms with suitable examples:

- (i) **Protonema:** Creeping, green, branched, filamentous juvenile stage in moss life cycle developing from spore (e.g., *Funaria*).
- (ii) **Antheridium:** Multicellular male sex organ producing biflagellate male gametes/antherozoids (e.g., Bryophytes/Pteridophytes).
- (iii) **Archegonium:** Flask-shaped female sex organ producing a single egg (e.g., *Marchantia*).
- (iv) **Diplontic:** Life cycle where the diploid sporophyte is dominant, photosynthetic, and independent (e.g., *Fucus*, Gymnosperms).
- (v) **Sporophyll:** Leaf-like structure bearing sporangia (e.g., Ferns, *Selaginella*).
- (vi) **Isogamy:** Fusion of gametes similar in size and motility (e.g., *Ulothrix, Spirogyra*).

Q9: Differentiate between the following:

- (i) **Red Algae vs Brown Algae:** Red algae contain r-phycoerythrin and store floridean starch; brown algae contain fucoxanthin and store laminarin/mannitol.
- (ii) **Liverworts vs Mosses:** Liverworts have dorsiventral thalli and simple spore dispersal; mosses have leafy upright axes and an elaborate spore dispersal mechanism with a protonema stage.
- (iii) **Homosporous vs Heterosporous Pteridophytes:** Homosporous produce one kind of spore; heterosporous produce microspores and megaspores.

Q10: Match the following (Column I with Column II):

- (a) Chlamydomonas → (iii) Algae
(b) Cycas → (iv) Gymnosperm
(c) Selaginella → (ii) Pteridophyte
(d) Sphagnum → (i) Moss

Q11: Describe the important characteristics of gymnosperms.

Answer:

1. Ovules are naked (not enclosed by ovary wall).
2. Plants are woody trees or shrubs with tap roots (Mycorrhiza in *Pinus*, Coralloid in *Cycas*).
3. Needle-like leaves in conifers with thick cuticle and sunken stomata adapt to extreme conditions.
4. Heterosporous: Form microspores and megaspores arranged in male and female cones/strobili.
5. Pollen tube delivers non-motile male gametes directly to archegonia in ovules.